Cell Biology Primer

A fictional reading for Biology 101 (Dr. Sam Okafor). Written for the Tramado demo.

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1. What a cell is

Every living thing is built from cells, the smallest units that can grow, use energy and copy themselves.

Some organisms are a single cell; a person is made of tens of trillions of them, working together.

Cells share a common plan: an outer boundary, a watery interior and instructions written in DNA.

2. The boundary of the cell

The plasma membrane separates the inside of the cell from its surroundings.

It is built from two layers of phospholipids, with their water-loving heads facing out and their tails facing in.

Proteins sit in or on this double layer and do most of the membrane’s work: carrying, signalling and anchoring.

3. Crossing the membrane

Small non-polar molecules such as oxygen and carbon dioxide diffuse directly across the lipid bilayer.

Ions and large polar molecules, such as glucose, need help from transport proteins.

Diffusion always runs down a concentration gradient and costs the cell no energy.

3.1 Channels and carriers

Channel proteins form narrow pores that let particular ions through very quickly.

Carrier proteins bind the molecule they move and change shape to release it on the other side.

3.2 Active transport

Pumps move substances against their gradient and pay for it with energy from ATP.

The sodium–potassium pump keeps the inside of animal cells low in sodium and high in potassium.

4. Water and osmosis

Osmosis is the diffusion of water across a membrane that lets water through but holds back the solutes.

In a hypotonic solution, water moves into the cell by osmosis and the cell swells.

In a hypertonic solution the opposite happens: water leaves and the cell shrinks.

4.1 Plant cells and turgor

A plant cell does not burst in pure water, because its cell wall pushes back as it fills.

This pressure, called turgor, keeps leaves and young stems firm.

5. Energy in the cell

Mitochondria release the energy stored in food and capture it in molecules of ATP.

Chloroplasts, found in plant cells, capture light energy and use it to build sugars.

5.1 Enzymes

Enzymes are proteins that speed up chemical reactions without being used up.

Each enzyme has an active site that fits its substrate closely.

6. Review questions

Why can oxygen enter a cell without a transport protein?

What would happen to a plant cell placed in salty water?

Which kind of transport needs ATP, and why?

Glossary

Bilayer: the double layer of lipids in a membrane. Gradient: a difference in concentration.

Hypotonic: having fewer dissolved particles than the cell. Turgor: the pressure of water inside a plant cell.